

Big Data Final Project

Weather Impact on Urban Traffic Analysis

Urban Traffic Analytics Under Weather Conditions

Goal:

Design and implement a **modern predictive data lake system** to analyze how **weather conditions affect urban traffic patterns**, using:

- **Synthetic Data**
- **MinIO (Bronze / Silver / Gold layers)**
- **HDFS (as an additional distributed storage layer)**
- **Python for data processing**
- **Monte Carlo Simulation**
- **Factor Analysis**

The final objective is to generate **actionable insights for urban traffic planning and traffic management under different weather scenarios**.

1. Project Scenario

A smart city authority in **London** wants to understand how **weather conditions (rain, temperature extremes, humidity, wind, visibility)** influence **traffic behavior and congestion levels**.

The authority provides **raw and messy weather and traffic datasets** collected from multiple sensors and monitoring systems. These datasets contain:

- Missing values
- Duplicated records
- Incorrect formats
- Outliers and extreme values

Your team is responsible for building a **full predictive data lake pipeline** that:

1. Stores raw data securely.
2. Cleans and structures the data.

3. Copies structured data to a distributed file system.
 4. Applies **Monte Carlo Simulation** to predict congestion risks.
 5. Applies **Factor Analysis** to identify the strongest weather drivers of traffic behavior.
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2. Dataset Description (Synthetic Data)

You will work with **two large synthetic datasets**:

1- Weather Dataset

Contains approximately **5000 records** representing:

- Different **times of day**
- Different **days**
- Different **seasons** (Winter, Spring, Summer, Autumn)

Main features include:

- Date & time
- City (single city: **London**)
- Season
- Temperature
- Humidity
- Rainfall
- Wind speed
- Visibility
- Weather condition
- Air pressure

The dataset also includes:

- Missing values
 - Outliers
 - Duplicates
 - Mixed date formats
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2- Traffic Dataset

Contains approximately **5000 records** representing traffic measurements at the same city and different times.

Main features include:

- Date & time
- City (same city: **London**)
- Area / district
- Vehicle count
- Average speed
- Accident count
- Congestion level
- Road condition
- Visibility

This dataset also contains messy real-world problems such as:

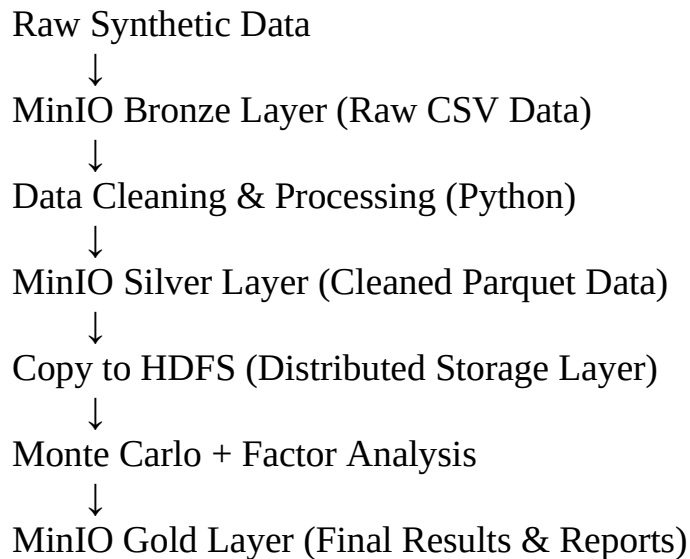
- Negative speeds
- Extreme vehicle counts
- Missing areas
- Incorrect congestion categories
- Duplicated rows

3. Technology Stack

Category	Tools
Infrastructure	Docker, MinIO, HDFS
Programming Language	Python
Data Formats	CSV (Raw), Parquet (Cleaned)
Analysis Methods	Monte Carlo Simulation, Factor Analysis
Visualization (Optional)	any Python dashboard tool

4. Data Lake Architecture

Your system follows a **three-layer data lake architecture using MinIO**, with an **additional HDFS integration layer**:



5. Project Phases (Step-by-Step)

Phase 1: Infrastructure & Data Ingestion (**Bronze Layer**)

Objective:

Set up MinIO and store the raw datasets.

Tasks:

1. Create a docker-compose.yml to start:
 - MinIO server
2. Create three MinIO buckets:
 - bronze → raw data
 - silver → cleaned data
 - gold → final results
3. Upload:
 - Raw weather dataset
 - Raw traffic datasetinto the **Bronze bucket**.

Deliverable:

- Screenshot of MinIO showing:

- The three buckets
 - Uploaded raw CSV files
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Phase 2: Data Cleaning (**Bronze** → **Silver**)

Objective:

Transform messy raw data into clean analytical datasets.

Tasks:

Clean both datasets by:

- Removing duplicates
- Handling missing values
- Fixing incorrect formats (dates, numbers, text)
- Removing or correcting outliers
- Standardizing category values

After cleaning:

- Save the cleaned datasets as **Parquet files**
- Store them in the **Silver bucket**

Deliverable:

- Cleaned Parquet files in the Silver layer
 - Data Quality Report:
 - Number of rows before cleaning
 - Number of rows after cleaning
 - Summary of what was fixed
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Phase 3: HDFS Integration (Additional Storage Layer)

Objective:

Integrate a **distributed file system** with the data lake.

Tasks:

- Copy cleaned Parquet files from:
 - **MinIO Silver** → **HDFS**
- Organize HDFS folders for:
 - Weather data
 - Traffic data

Deliverable:

- Screenshot of HDFS showing uploaded Parquet files
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Phase 4: Dataset Merging

Objective:

Create a unified dataset for analysis.

Tasks:

- Merge weather and traffic cleaned datasets using:
 - date_time
 - city
- Produce a single **merged analytical dataset**

Deliverable:

- Final merged dataset stored in Silver or Gold layer
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Phase 5: Monte Carlo Simulation (Traffic Risk Prediction)

Objective:

Simulate traffic behavior under different weather conditions.

Simulation Scenarios Must Include:

- Heavy rain
- Temperature extremes
- High humidity
- Low visibility
- Strong winds

Main Outputs:

- Probability of:
 - Traffic jams
 - Accident risk under bad weather
- Distribution of congestion probabilities from thousands of simulation runs

Deliverable:

- simulation_results.csv
 - Congestion probability distribution plot
Saved in **Gold layer**
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Phase 6: Factor Analysis (Weather Impact Detection)

Objective:

Identify which **weather variables** have the **strongest effect on traffic patterns**.

Input Features Should Include:

From Weather:

- Temperature
- Humidity
- Rain
- Wind speed
- Visibility
- Air pressure

From Traffic:

- Vehicle count
- Average speed
- Accident count

Expected Output:

- 1–3 hidden latent factors such as:
 - Weather Severity Factor
 - Traffic Flow Stress Factor
 - Accident Risk Factor

Deliverable:

- Factor Loadings Table
 - Interpretation Report:
 - Which weather variables affect traffic the most
Saved in the **Gold layer**
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Phase 7 (Optional Bonus): Visualization Dashboard

Objective:

Build a simple interactive web dashboard.

Dashboard Must Display:

- Cleaned dataset statistics
- Monte Carlo simulation results
- Factor Analysis insights

Tools:

Streamlit or any Python-based visualization tool.

Final Project Deliverables

Students must submit:

1. Synthetic raw weather & traffic datasets
 2. Cleaned Parquet datasets in Silver
 3. Screenshots of:
 - MinIO Bronze, Silver, Gold buckets
 - HDFS storage
 4. Merged dataset
 5. Monte Carlo simulation results
 6. Factor Analysis report
 7. Optional dashboard
 8. Final Technical Report (PDF) including:
 - Data cleaning summary
 - Simulation methodology
 - Factor interpretation
 - Final insights & recommendations
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Team Members : 5 – 10 members

Deadline : 14/12 at 12 midnight

Synthetic Weather Dataset — Rules of Generated Data

Column Name	Data Type (Expected)	Description	Possible Values & Messy Scenarios (Included)
weather_id	Integer	Unique ID for each weather record	- Standard: 5001, 5002, 5003... - Duplicates: same ID appears in multiple rows - Some rows with missing ID (NULL)
date_time	DateTime	Timestamp of the weather measurement	- Standard: 2024-01-15 14:00 (YYYY-MM-DD HH:MM) - Format variations: 15/01/2024 2PM, 2024-01-15T14:00Z - Invalid / garbage: 2099-13-40 25:61, Unknown - Some NULL timestamps
city	String	City name (single city used for the project)	- Standard: London - Some NULL city values
season	Category	Season of the year (derived from date_time)	- Standard: Winter, Spring, Summer, Autumn - Some NULL or inconsistent season entries due to bad date formats
temperature_c	Float	Measured temperature in Celsius	- Expected realistic range by season (e.g., Winter: -5–15, Summer: 10–35) - Outliers included: -30, 60 - Some NULLs
humidity	Integer (%)	Relative humidity percentage	- Expected: 20–100 - Outliers included: -10, 150 - Some NULLs
rain_mm	Float	Rainfall amount in millimeters	- Typical: 0–50 - Extreme values included: 120+ - Some NULLs
wind_speed_kmh	Float	Wind speed in km/h	- Typical: 0–80 - Outliers included: 200+ - Some NULLs
visibility_m	Integer	Visibility in meters	- Typical: 50–10,000 - Extreme: 50,000 - Some NULLs and non-numeric strings in a small fraction
weather_condition	Category	Weather type label	- Accepted: Clear, Rain, Fog, Storm, Snow - Some NULLs

air_pressure_hpa	Float	Atmospheric pressure in hPa	- Typical: 950–1050 - Some NULLs
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Synthetic Traffic Dataset — Rules of Generated Data

Column Name	Data Type (Expected)	Description	Possible Values & Messy Scenarios (Included)
traffic_id	Integer	Unique ID for each traffic record	- Standard: 9001, 9002, 9003... - Duplicates present - Some NULL IDs
date_time	DateTime	Timestamp of traffic measurement	- Standard: 2024-01-15 08:00 - Format variations: 15/01/2024 8AM, 2024-01-15T08:00Z - Invalid / garbage: TBD, 2099-00-00 99:99 - Some NULLs
city	String	City name	- Standard: London - Some NULLs
area	String	District / area name	- Examples: Camden, Chelsea, Islington, Southwark, Kensington - Some NULL area values
vehicle_count	Integer	Number of vehicles recorded	- Expected: 0–5,000 - Extreme outliers included: 20,000+ - Some NULLs
avg_speed_kmh	Float	Average vehicle speed (km/h)	- Expected: 3–120 - Invalid values included: negative speeds - Some NULLs
accident_count	Integer	Number of accidents during measurement interval	- Expected: 0–10 - Rare extreme values included: 50+ - Some NULLs
congestion_level	Category	Low / Medium / High	- Standard categories present - Some NULLs
road_condition	Category	Road surface condition	- Standard: Dry, Wet, Snowy, Damaged - Some NULLs
visibility_m	Integer	Visibility in meters (traffic sensor)	- Typical: 50–10,000 - Extreme values and some NULLs